Frederick Huang

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RESEARCH INTERESTS

Design, synthesis, and mesoscale modeling of functional polymeric materials using machine learning (ML) frameworks for accelerating materials discovery, paired with experimental approaches. Broad topics of interest include data-driven/classical modeling, inverse design, and autonomous experimentation. Current work is in development of antimicrobial, polymeric composites, validation of ML predictions, and literature mining for legacy materials data.

EDUCATION

University of North Carolina at Chapel Hill and North Carolina State University 2025
B. S., *magna cum laude,* in Biomedical Engineering, minor in Chemistry

RESEARCH EXPERIENCE

Research Aide, Argonne National Laboratory January 2027–present
Applied Materials Division | Advisor: Wenhua Yu, Ph. D.

- Composite base fluids for large power transformers to improve alternating current and impulse breakdown voltages.

Research Assistant, University of Texas Health Science Center at San Antonio November 2025–December 2025
Department of Comprehensive Dentistry | Advisor: Kyumin Whang, Ph. D.

- Built a literature-mining pipeline to compile multimodal data, generating a structured dental composites dataset of ~290 unique formulations from ~30 experimental sources to expand model training data and open-source its use.
- Leveraged Scopus/OpenAlex API for mass query and download of relevant experimental dental materials publications.
- Developed quaternary ammonium silane functionalized polymer composites incorporated with phosphate ceramics to demonstrate antifouling activity against *S. mutans*, *S. sanguinis*, and *C. albicans* with remineralization capability in clinical restorations.
- Synthesized and characterized polymer composite formulations predicted by a collaborator-developed classical machine learning model using mechanical load frames, FT-IR, colorimetry, etc. within ISO 4049 standards to validate predicted performance and inform model refinement.
- Designed and built a custom curing apparatus, reducing degree of conversion variability in batch sample production.
- Conducted blinded testing on industry-sponsored projects, identifying batch-to-batch manufacturing inconsistencies and reporting findings to production teams for quality review.

This work resulted in 2 submitted abstracts (first and second author), funded in part by Pulpdent, Inc.

Undergraduate Research Assistant, UNC-Chapel Hill August 2024–May 2025
Joint Department of Biomedical Engineering | Advisor: Jeffrey Macdonald, Ph. D.

- Analyzed heterogeneous LC-MALDI-TOF lipidomic datasets from *R. norvegicus* brain and *E. fuscus* liver tissue using LipidSearch 5, identifying phosphatidylcholine and phosphatidylethanolamine as lipid classes most sensitive to sample preparation conditions.
- Built quantitative visualizations to evaluate data quality across preparation conditions, isolating lipid features with statistically significant degradation signatures.
- Proposed candidate lipids in specimens as features for a lipidomic integrity number (LIN), a data quality metric addressing a gap left by existing RNA/DNA-based quality control standards (RIN, DIN).

This work resulted in 1 abstract, funded by NCSU/UNC's Abrams Scholarship program.

Undergraduate Research Assistant, UNC-Chapel Hill December 2023–August 2024
Division of Pharmacoengineering | Advisor: Zhenzhen Chen, Ph. D.

- Synthesized liposomal drug carriers via thin-film hydration and rotary evaporation, followed by extrusion for size control,

and prepared metal nanoparticles for transmission electron microscopy (TEM) imaging to characterize morphology.

TEACHING EXPERIENCE

- **BMME209**, *Materials Science of Biomaterials*, Teaching Assistant, Spring 2025

POSTERS / TALKS

1. Khan, M. S., **Huang, F.**, Whang, K. (2027, submitted). *Effects of Quaternary Ammonium-Modified Bioactive Fillers on Composite Properties*, American Association for Dental, Oral, and Craniofacial Research and Canadian Association for Dental Research Annual Meeting (AADOCR/CADR), Minneapolis MN.
2. **Huang, F.**, Whang, K. (2027, submitted). *AI-Based, Automated Extraction of Dental Materials Data from Literature*, American Association for Dental, Oral, and Craniofacial Research and Canadian Association for Dental Research (AADOCR/CADR) Annual Meeting, Minneapolis MN.
3. **Huang, F.**, Arciniega C., Ehrmann B., Macdonald, J. (2025). *Quality Management in Lipidomic Extractions to Develop a Lipidomic Integrity Number*, UNC-Chapel Hill Undergraduate Research Symposium, Chapel Hill NC.
4. Mathur, S., Paye, K. S., **Huang, F.**, Cole, M. N., Gilbert, M. J., Hederick, M. K. (2025). *Brace Check—Data Collection and Monitoring for Pectus Carinatum Patients*, Joint Department of Biomedical Engineering Senior Design Symposium, Raleigh NC.

HONORS & AWARDS

Interprofessional Practice and Education Award, *UNC-Chapel Hill* 2025
Awarded through School of Medicine for capstone design project addressing pediatric patient needs in physical medicine and rehabilitation.

Abrams Scholarship, *North Carolina State University* 2024
Awarded through College of Engineering to 30-35 Biomedical Engineering students to support research under faculty guidance.

COMMUNITY SERVICE & LEADERSHIP

Classical Music Instructor, *Musical Empowerment*, UNC-Chapel Hill 2022–2025
Galamian-based curriculum for beginner-intermediate violinists from underserved backgrounds within Chapel Hill-Carrboro City Schools District.

Vice President, *Club Ultimate Frisbee*, UNC-Chapel Hill 2022–2025
Finance and Operations, >$60,000 yearly operating budget.

PROFESSIONAL EXPERIENCE

IP/Technology Transfer Intern, UT Health San Antonio 2026

SKILLS AND TECHNICAL COMPETENCIES

- **Characterization:** FT-IR, SEM, DSC, DMA, Contact Angle, Indentation, Mechanical Load Frames
- **Packages:** Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly, SciPy
- **Tools:** Jupyter, PlatformIO, Git
- **Software:** Onshape, Fusion360, KiCAD, PyMOL, Chimera
- **Miscellaneous:** Sewing, Sectioning/microtomes, Soldering, Microcontrollers, Linux, LaTeX